FICTION

Popular lessons in astronomy, on a new plan : in which some of the leading principles of the

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POPULAR

LESSONS IN ASTRONOMY,

ON A NEW PLAN;

IN

WHICH SOME OF THE LEADING PRINCIPLES OF THE SCIENCE ARE ILLUSTRATED BY ACTUAL COMPARISONS, INDEPENDENT OF THE USE OF NUMBERS.

BY

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[Illustration: Decoration]

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NOTICE.

The Geographical Miles, which are used in the scales and tables of this book, are _German Geographical Miles_, of which 15 make one degree. The teacher or pupil may easily change them into _English Geographical_ Miles, by multiplying them by 4, the square miles by 16, and the cubic miles by 64.

PREFACE.

Although many elementary works on Astronomy are already before the public, yet it is believed there is none in which the various magnitudes, surfaces, and distances of the heavenly bodies, are presented to the eye of the learner by actual comparisons;—the only way, perhaps, in which young pupils can be expected to form a correct idea of them. This the author has attempted in the following pages. The comparative diameters, surfaces and distances of the different planets, are all drawn,

in the plates, according to a fixed scale of geographical miles; the surfaces of the planets are actually reduced to square measure, and drawn in proportion to each other and the sun; so that the youngest pupil, by a mere glance of the eye, is enabled to form a correct idea of their respective magnitudes. A similar plan has been pursued, in regard to the division of the Earth into Zones, and with respect to the extent of the five great Continents of our Globe. The Appendix contains an exposition of the population of America, Europe, Asia, Africa, and Australia, accompanied by a plate for the illustration of the comparative settlements on those Continents.

Boston, June 24, 1833.

POPULAR LESSONS IN ASTRONOMY.

LESSON I.

THE EARTH IN ITS RELATION TO THE SUN, MOON AND STARS.

§ 1. The Earth on which we live, and on which plants, trees and animals successively live and die, is only a small part of the world; it is but one of the smallest bodies in the Universe. To the world belong yet the _Sun_ from which we receive warmth and light, the _Moon_, and an innumerable class of bodies, which, at night, appear to us as so many points of light. These are called _Stars_.

The reason why the Stars appear to us so small, is because they are so far from us; and things appear smaller in proportion as they are farther removed from us. This you will have noticed, when looking from a high steeple on the people below, or on a vessel far out in the harbour, or on a chain of mountains at a great distance.

§ 2. Most all Stars appear to us, every night in the same position; they seem actually to be _fixed_ in the heavens; and for this reason they are called _fixed Stars_. There are however Ten others, of which it has been ascertained _that they move regularly round the Sun in large circles_. These are called _Planets_ or _wandering Stars_.—The fixed Stars are supposed to be similar to the Sun, in as much as they are bodies which have their own light. The Planets, on the contrary, are of themselves dark bodies, and receive, like our Earth, light and warmth from the Sun. We see them only in consequence of the solar light which they _reflect_ from their surfaces, and this is the reason why they appear to us as bright as the other Stars.

§ 3. The discoveries of Philosophers have proved beyond a doubt, _that our Earth itself is one of those Planets_, which move round the Sun in stupendous large circles, whose grandeur is hardly conceived by the most powerful imagination. _Our Earth, therefore, is, itself, a Wandering Star_, and the line in which it moves round the Sun is called its _Way_ or _Orbit_.

§ 4. The Planets, together with our Earth keep each a certain fixed distance from the Sun. On this account they do not disturb each other in their orbits. But they vary from each other in magnitude; although all of them (consequently also our Earth) have a _round_ shape, similar to a ball.

In former times men believed that the Earth was flat, or a circular plate on all sides surrounded by water. But this is not true. For it has been proved by a great many

observations and actual measurements, that the Earth has a spherical (ball-like) form. Moreover there are navigators who have actually _sailed round the world_, and who have noticed the fact that at sea, the tops of distant objects are seen sooner than the rest; which again proves the spherical form of our Earth. For an illustration you may look at the adjoining plate Plate No. I.

The man who is represented as standing on a portion of our Earth, will at first only see the topmast of a vessel at sea; when she comes nearer his eye will discover a much greater portion of her; but when in the third position every part of the vessel will be visible.

§ 5. Some of the Planets are, in their motion round the Sun, _accompanied by other dark bodies_, which, like the Planets themselves, receive light and heat from the Sun. These are called _Satellites_ or _Moons_. Such a Satellite is the _Moon which accompanies our Earth_; and there are Planets (as we shall see hereafter) which have Four and Six, and One that has even Seven Moons.

§ 6. Besides the Planets and Moons there is yet another class of bodies, moving round the Sun in exceedingly long ovals. They are but seldom visible, and are distinguished from other heavenly bodies by a tail which is often three, four, and more times larger than the body itself. These are called _Comets_; but their number has not, as yet, been precisely ascertained.

§ 7. The Planets, Moons and Comets, together with the Sun around which they move, form what is called our _Solar System_. But what will you say, if you are told that each fixed Star in the firmament is a Sun which, like our’s, has its Planets and satellites and Comets; in short, that each fixed Star is the centre of a solar system, a thousand and more times larger than our own!! But how infinitely great must God’s Creation appear to us, when we reflect that all these globes, as well as our own, may be inhabited by reasonable beings!!!

[Illustration: _N^o. I._]

RECAPITULATION OF LESSON I.

[If the pupil has learned and understood this lesson, it may perhaps be not unreasonable to suppose he will like to know something more about the Planets and the Moons in our system. But the teacher ought not to allow him to enter upon the second lesson, before he has recapitulated the first.]

QUESTIONS.

[§ 1.] Does the Earth, or the globe on which we live, comprise the whole world which God has created? Is our Earth any considerable portion of the Universe? What other bodies belong to it? Why do the Stars appear to us so small?

[§ 2.] Do all the Stars which we see remain in the same position? What are those called which remain _fixed_? What, those, which are moving regularly round the Sun? What are the fixed Stars supposed to be? Have the Planets any light of their own? From what body do the Planets receive their light?

[§ 3.] What important discoveries have Philosophers made respecting the nature of our Earth?

What is the line in which our Earth moves round the Sun, called?

[§ 4.] Why do the orbits of the different Planets not disturb each other? Are all Planets of the same magnitude? What is their shape? What is the shape of our Earth? What reason have you to believe, that the Earth is round? Explain Plate I?

[§ 5.] By what are some Planets in their motion round the Sun accompanied? What are the bodies which accompany them called? What body accompanies our Earth in its motion round the Sun? Are there Planets which have more than One Moon?

[§ 6.] What other class of bodies is there, besides the Planets and the Moons? Are these bodies always visible? By what are these bodies distinguished? What are they called?

[§ 7.] What do the Planets, Moons and Comets, together with the Sun, round which they move, form? What is each fixed Star supposed to be?

LESSON II.

OF THE PLANETS AND THEIR RELATIVE POSITION WITH REGARD TO THE SUN AND EACH OTHER.

§ 8. The Planets of which we have spoken in the First Lesson are Eleven in number, of which each has received a proper name. They are in the order in which they are placed from the Sun, the following:

Mercury. Venus. The Earth. Mars. Vesta. Juno. Ceres. Pallas. Jupiter. Saturn. Herschel.

The adjoining diagram, on Plate No. II, may serve for an illustration. You will see from it, that the Earth is the third Planet from the Sun. Mercury and Venus come before it; the most remote from the Sun is _Herschel_, so called from the astronomer’s name, who has but lately discovered it.

§ 9. The Earth is constantly attended by One Satellite (Lesson I, § 5); Jupiter by Four (see the diagram), Saturn by Six, and Herschel by Seven. Saturn is, besides, constantly surrounded by a broad luminous ring, which distinguishes it from all other Planets, and of which we shall speak hereafter.

§ 10. With regard to the magnitude of the different Planets it has been observed that although some of them are much larger than Our Earth; yet in Comparison to the Sun they are but small bodies, as you may see from the figures on plate No. III, in which the different Planets are drawn in proportion to the Sun: you will perceive from it;

1. That the Sun is a great many times[1] larger than either of the Planets.

[Illustration:

_N^o. II._]

[Illustration:

N°. III..]

2. That the Sun is much larger than all the Planets (our Earth included) taken together.

3. That Jupiter is the greatest Planet in our solar system.

4. That Saturn is the second Planet with regard to magnitude.

5. That Herschel ranks third.

6. That the Earth ranks fourth; but that

7. Each of the other Planets and the Moon are smaller than the Earth.

8. That either of the Planets Ceres, Vesta, Juno, Pallas is smaller than the Moon.

§ 11. If you cut a ball or a sphere in halves, and through the centre of one of the flat surfaces, draw a straight line across the whole surface, then this line is called the _Diameter_ of the ball or sphere.—Now as the Sun and the Planets are nothing else but large balls or spheres (Lesson I, § 4), we may also speak of their Diameters.—Thus we say, “the Diameter of the Sun is so many times larger than that of the Earth; the Diameter of Jupiter is so many times larger than that of Venus,” etc.

You will now easily understand what the straight line means, which, in plate No. III is drawn through the centre of the circle, which represents the Sun; also the straight lines which are drawn through the circles which represent the Planets.—You will also perceive from the first figure on plate No. III, that the Sun’s Diameter is larger than the Diameters of all the Planets taken together.

§ 12. Plate No. IV represents the _surfaces_ of the Planets, compared to that of the Sun.—For you must know that it is possible to _measure_ the surface of a sphere; which is done by supposing it to be spread out, and then seeing how many square inches, feet or yards it contains. This we have supposed to be done with the surfaces of the Sun and Planets, and accordingly, have drawn their square measurements, in Plate No. IV. in proportion to each other.

You will see from this Plate:

1. That the square-Contents of the Sun’s surface is greater than that of the surfaces of all the Planets taken together.

2. That the square-Contents of the Earth’s surface is nearly as large as that of the surfaces of Mercury and Venus taken together.

[Illustration:

_N^o. IV._]

3. That the square-Contents of the surfaces of Vesta, Ceres, Juno and Pallas, taken together, is not yet sufficient to cover the surface of our Earth.

4. That the square-Contents of Jupiter’s surface is larger than that of the surface of any other Planet.

5. That the square-Contents of Jupiter’s surface is greater than that of the surfaces of all other Planets taken together.

6. That Saturn’s surface is the next largest.

7. That Herschel’s is still larger than those of our Earth, Venus and Mercury taken together.

8. That the square-Contents of the Moon is nearly as large as that of the surfaces of Ceres, Vesta and Juno, taken together; and that it is larger than that of Pallas’ surface taken alone.

In order that you may be better able to compare, you will find on Plate No. V

1. The surfaces of the different Planets, all drawn upon the Sun’s surface; from which you may see, how much larger the Sun’s surface is than the surfaces of all the Planets and the Moon taken together.

2. The surface of the Earth compared to that of Mercury and Venus.

3. The Earth’s surface compared to that of the Four Asteroids Pallas, Ceres, Juno and Vesta.

4. Jupiter’s surface compared to those of all the other Planets.

5. The square-Contents of the Moon compared to the surfaces of Vesta, Juno and Ceres.

6. Herschel’s square-Contents compared to that of our Earth, Venus and Mercury taken together.

§ 13. The Four Planets, Ceres, Vesta, Juno, and Pallas, are so small in proportion to the other Planets, and are so near each other (they are almost at the same distance from the Sun—see Plate II) that many distinguished philosophers are of opinion, they are but the fragments of One large Planet, which, from some cause or other, has _burst_.—But this is no subject for us now to inquire into;—we will therefore proceed to describe in the next lesson, the motion of the different Planets round the Sun.

[Illustration:

_N^o. V._]

RECAPITULATION OF LESSON II.

QUESTIONS.

[§ 8.] What are the names of the Eleven Planets _in the order in which they are placed from the Sun_? How many Planets are there between the Earth and the Sun? How many Planets come after our Earth? In what order then, is our Earth placed among the Planets? (Is it the first, second, etc. from the Sun)?

[§ 9.] How many satellites accompany our Earth? What is it called? How many satellites or Moons has Jupiter? How many has Saturn? How many has Herschel? By what is Saturn particularly distinguished from all other Planets?

[§ 10.] Are all Planets of the same magnitude as our Earth? Are the Planets in comparison to the Sun large or small bodies?—Is the Sun much larger than any of the Planets? Is the Sun larger or smaller than all the Planets, including our Earth, taken together? Which is the greatest of all Planets? Which Planet ranks, in magnitude, next to Jupiter? Which of the Planets ranks third in magnitude? Which, Fourth? What Planets are smaller than our Earth? What Planets are smaller than the Moon?

[§ 11.] What do you call the _Diameter_ of a ball or sphere? Is the Diameter of the Sun larger or smaller than the Diameters of all the Planets taken together?

The teacher may yet ask a number of questions respecting the Diameters of the Planets, which the pupil will be able to answer by looking on Plate III. He may, for instance, ask whether the Diameter of the Earth is larger or smaller than that of Venus? Whether the Diameter of Jupiter or Saturn is the largest, etc.? This will oblige the pupil to _compare_ the different magnitudes of the Planets and their Diameters.

[§ 12.] What proportion does the square-Contents of the Sun’s surface bear to that of the surfaces of all the Planets? What two Planets have together a surface nearly as large as that of our Earth? Are the surfaces of Ceres, Vesta, Juno and Pallas taken together sufficient or not, to cover the surface of our Earth? What Planets, therefore, could be covered with the Earth’s surface? Which Planet has the largest surface? What do you observe with regard to the square-Contents of Jupiter’s surface? and that of the surfaces of all other Planets, taken together? What Planet’s surface comes next in magnitude to Jupiter’s? What Planet’s surface is larger than that of our Earth, Venus and Mercury taken together? What three Planets’ surfaces taken together are smaller than that of the Moon? What Planet has nearly as large a surface as the Moon?

[§ 13.] What is the opinion of philosophers respecting the Four Planets, Ceres, Vesta, Juno and Pallas?

FOOTNOTES:

[1] Numbers are purposely omitted here, because abstract numbers convey little or no ideas to young pupils. Those who wish these relations expressed in numbers will find them in Table I at the end of the book.

LESSON III.

EXPLAINING THE MOTION OF THE PLANETS AND COMETS ROUND THE SUN.

§ 14. You have learned, in the preceding lesson, that the Earth and the other Planets are regularly moving round the Sun; but you must know that the time which each requires to complete a whole revolution, that is, the time which each needs to move once round, cannot be the same with all; as you may easily judge yourself: for the Planets which are next to the Sun will, of course, have a much smaller journey to perform, than those which are further from it. Thus Mercury, which is the first Planet in order from the Sun, will naturally come round much sooner than Jupiter, which is placed at a much greater distance from it.

That you may the easier understand this, the following Plate, No. VI, will represent to you our Solar System:

The Sun from which proceeds all light and heat, is placed in the centre. Then come the Planets Mercury and Venus; Third in order is our Earth with its Satellite the Moon; and so on. The rest of the Planets in the same order in which they are represented on Plate II. You will also perceive there the orbs of two Comets, distinguished as you were told, by a tail of light. The Planets Jupiter, Saturn and Herschel are each represented with their Moons, and the Planet Saturn with its luminous ring.

[Illustration:

_N^o. VI._]

Now you will easily understand that Ceres, Vesta, Juno, Pallas, Mars, Jupiter, Saturn and Herschel, have each a much greater distance to travel, than our Earth; but that, on the contrary, Venus and Mercury can complete their revolution in a much shorter time.[2]

§ 15. The time which our Earth needs to travel once completely round the Sun, (to finish one revolution) is called a _Year_. Such a year has three hundred and Sixtyfive days;—and each of these days has again Twentyfour hours.

You see from this that the revolution of the Earth round the Sun gives us _a means of measuring time_, by which we are able to bring order and regularity into our business and transactions of life. For the making of clocks and watches is but a late invention, and we should be left entirely in the dark as regards the history of former ages, and a great many people would, at this present moment, be incapable of forming a correct estimate of time, if Providence had not given to all this appropriate means of measuring it.

§ 16. While the Earth needs a whole year for one revolution round the Sun; Mercury requires but Eightyone days, and Venus only about two thirds of One of our years. Mars, on the contrary, needs for one of his revolutions almost Two years; Vesta almost Four; Juno, Ceres and Pallas over Four years; Jupiter almost Twelve, Saturn over Twentynine, and Herschel nearly Eightyfour of our years![3]—And if these Planets, as we have reason to believe, are inhabited by beings endowed with human understanding and faculties, numbering their years as we do ours—by the revolution of their Planets round the Sun—how different from ours must be the Period of their existence!!

§ 17. During the time that the Earth is performing her journey round the Sun, the Moon, our constant attendant, is continually moving round the Earth, _and completes one of these revolutions in little more than Twentyseven days_.—Very important and interesting to us are the changes in appearance which she exhibits during each of these revolutions.—You probably will know, that the Moon does not appear to us, at all times, the same. Sometimes she is hardly at all visible, (at least not with the naked eye); at other times only a small rim of her is seen, which by degrees becomes larger and larger, until finally she appears in her full round form. After this she begins again

to diminish, changes again into a small luminous rim, and finally disappears entirely from our sight. These successive changes in the Moon's appearance are called _the Moon's Phases_, or the _waxing and waning of the Moon_. The time during which the Moon is not seen is called _New Moon_; the time during which she exhibits her full shape is called the _Full Moon_; and the different periods of her waxing and waning (when she appears to us in the form of a crescent) are called _Quarters_. Thus we speak of the _First_ and of the _Last_ Quarter of the Moon. The First Quarter takes place after New Moon; the last Quarter after Full Moon.

The following diagram, Plate No. VII, may serve to represent to you the Moon's phases as seen from our Earth.

When the Moon is in _a_, then the light of the Sun falls just on that side of it which is turned from the Earth. It is then, we have New Moon. When in _b_, a small brim of the Moon is seen, because a small portion of its lighted surface is then turned towards the Earth.—When in _c_ half of her lighted surface is turned towards us, and we have the First Quarter. In _d_ a still greater portion of the Moon's lighted surface is visible, and in _e_, we have Full Moon, because her whole lighted surface is then turned towards the Earth. In _f_ the moon commences to wane (to grow smaller,) and in _g_ the last quarter commences; finally, when passed through the point _h_, we have in _a_, again New Moon. For familiar illustration you may also take a white ivory ball, holding it before a lighted candle, which may take the place of the Sun. When the ball is in a straight line between your eye and the candle it will appear to you all dark; because the lighted part is then entirely turned toward the candle (away from you), and you have the same case which is represented to you in the diagram, when the Moon is in _a_. But if you move the ball a little to the right, you will perceive a streak of light, similar to the First Quarter represented in the Diagram, when the Moon is in _c_. If moved still farther to the right, so that the whole lighted part of the ball is seen, it will resemble the Full Moon; represented in the Diagram, when the Moon is in _e_.

[Illustration:

N°. VII.]

[Illustration:

N°. VIII.]

§ 18. While the Moon is moving round the Earth, it often occurs that she is placed in a direct line between ourselves and the Sun. In this case a greater or less part of the Sun is concealed from us, which causes a diminution of light or a partial darkness on our Earth. This we call an _Eclipse of the Sun_. (Such an Eclipse took place in 1831, and you will probably have an opportunity of seeing many more). If, on the contrary, the Earth is placed in a direct line between the Sun and the Moon, then the Moon will be obscured by our Earth. This is called an _Eclipse of the Moon_. The following two figures on Plate No. VIII will serve for an illustration.

You will easily perceive from them that if the Moon (as represented in Figure I) is placed in a direct line between the Sun and ourselves, it must necessarily conceal

from us part of that luminary; and in this state cast a _shade_ upon our Earth.

But if the _Earth_ is placed in a direct line between the Sun and the Moon, (as represented in Figure II), then the Moon will be much more obscured, because the Earth is much larger than the Moon, and will therefore cast a much greater shade upon her.

§ 19. It remains for us to speak of that class of bodies known by the name of _Comets_, (see Lesson I, § 6). Of these an unknown number belongs to our Solar System.—(Some philosophers have estimated their number to be about Twentyone; others think it must amount to several hundred). They move round the Sun in exceedingly long ovals, having their transparent tails always turned away from that luminary. What is most remarkable about them is the astonishing degree of heat to which they are exposed on account of passing so near the Sun, and the astonishing velocity with which they travel.

The Comet which appeared in the year 1680, is supposed to sustain a heat nearly Two Thousand times greater than that of red hot iron, and to move at the rate of several Hundred Thousand miles an hour!!

RECAPITULATION OF LESSON III.

QUESTIONS.

[§ 14.] Do all Planets need the same time to complete a whole revolution round the Sun? Why not?

If the pupils are old enough to understand the use of Dividers, it will perhaps be well for the teacher to let them draw the Solar System on a piece of paper.—If not, he ought to let them explain Plate IV, or an orrery, if one be at hand.

[§ 15.] What is the time called, which our Earth needs for a complete revolution round the Sun? How many days are there in a year? How many hours are there in a day?

What is the revolution of the Earth round the Sun, the means of?

[§ 16.] What time does Mercury require for a complete revolution round the Sun? What time does Venus require for the same purpose? What time does Ceres, Vesta, Juno and Pallas need? What, Jupiter, Saturn, and Herschel?

[§ 17.] What motion does the Moon make whilst the Earth is travelling round the Sun? How many days does the Moon need for a complete revolution round the Earth? Does the Moon during its revolution round the Earth always exhibit the same shape? What changes then does she gradually undergo? What do you call the successive changes in the Moon’s appearance? What do you call the time during which the Moon is not seen? What, that, during which she exhibits her full shape? What, the different periods of her waxing and waning? When does the First Quarter take place? When the last?

[The teacher might now require the explanation of Plate V; the elder pupils may draw the Diagram.]

[§ 18.] What does frequently occur during the Moon’s motion round the Earth? What is the consequence of the Moon’s position in a direct line between the Sun and ourselves? What is such a diminution of light in consequence of the Moon being placed between us and the Sun, called? What takes place when the Earth is placed between the Sun and the Moon? What is such an obscuration of the Moon, in consequence of the Earth’s position between her and the Sun, called?

[The younger pupils ought now to explain Plate VIII; the elder pupils ought to draw the Diagram on a slate or paper.]

[§ 19.] Is the number of Comets belonging to our Solar System precisely ascertained? How many are there _supposed_ to belong to our System? In what manner do they move round the Sun? What remarkable property do they possess?

FOOTNOTES:

[2] If the teacher has an orrery at hand, if it be even of the most simple construction, he may exhibit it now. But it is the author’s belief that a considerable portion of the pupil’s interest is lost, if he be acquainted with it, at the beginning of the study; or, as it is the custom in some schools, if an orrery is hung up among the charts and maps of the school room. The pupil ought not to see the orrery, until he knows that it is but a _faint illustration_ of the infinite grandeur of the heavens. Nothing detracts so much from our estimation of things as a too familiar acquaintance with them, before we know their real value.

[3] The exact numbers are given in Table II, at the end of the book.

LESSON IV.

ROTATION OF THE SUN, THE EARTH, AND THE REST OF THE PLANETS ON THEIR AXES.—DAY AND NIGHT.—INCLINATION OF THE EARTH’S AXIS.—SEASONS.

§ 20. Besides the progressive motion of the Planets, of which we have spoken in the last Lesson, they have yet a peculiar motion like a wheel turning on its Axle-tree. This motion is called the _rotation of the Planets on their Axes_; because each of them seems to move round a straight line passing through its centre; like a ball turning round a piece of wire run through the middle of it.—Moreover it is customary to call such a straight line imagined to be drawn through the centre of a Planet—_the Axis of that Planet_.

The double motion of the Planets,—progressive and rotary,—is perhaps one of the most difficult things for young pupils properly to understand, without some popular illustration. If the teacher, therefore, has no orrery to show this motion to his pupils, he may compare it to a screw which is turned round whilst it suffers at the same time a progressive motion; or perhaps with more propriety to a spinning-top, which is continually turning on its Axis, while at the same time it describes large circles.[4]

§ 21. If you have well understood what has just been said, you will be able to comprehend, that _our Earth, while it is performing its great journey round the Sun

in Three Hundred and Sixtyfive days, is, at the same time, every Twentyfour hours turning on its Axis_. This _rotary_ motion of our Earth on its Axis is the cause of the successive changes of day and night; that portion of the Earth which is turned toward the Sun having always day, when the other, which is turned away from him, has night.

This is again a wise dispensation of God's providence. For if the Earth would always keep the same relative position to the Sun, then that portion of it, which would then be turned toward the Sun, would have continual day, whilst the other, which would then be turned away from him, would be enveloped in perpetual darkness. But as it is now arranged by the Earth's rotation on its Axis, most every portion of its surface must at least _once_ every Twentyfour hours be turned toward the Sun and receive from him light and heat. Without this, one great half of our Earth would have a perpetual winter, destructive to plants and animals, while an everlasting summer would scorch the other half and render it equally unfit for the support of man.

The following Diagram, Plate IX, may serve to give you an idea of the Earth's rotation round its Axis, and the alternate succession of Day and Night, resulting from it. When the Earth is situated as represented in the Diagram, then that portion of it, which is marked A, will have Day, because it is turned toward the Sun; and the portion marked B, will have Night. But in the course of the next Twelve hours the order will be reversed. The portion which is marked B, will be turned toward the Sun and have day, whilst the portion A will be turned from him, and have night.

§ 22. The rotation of the Earth on its Axis is also the cause of the _Rising_ and _Setting of the Sun_. For no portion of our Earth is at once turned toward or from the Sun; but moves toward or from it by degrees (as you may see by slowly turning a ball near the flame of a candle). This gradual motion of each portion of the Earth's surface toward or from the Sun, makes the Sun himself appear to us as rising and going down; while, in fact, we ourselves are turning towards, or receding from him. This is a kind of deception similar to that which you experience when slowly gliding down a river; when the objects on shore have the appearance of receding from you, while in fact, it is you, yourself, who are travelling away from them.

[Illustration:

_N^o. IX._]

The gradual rise and setting of the Sun is another excellent provision of nature. Were we from the darkness of night at _once_ exposed to the luminous rays of the Sun, it would dazzle our eyes and render them unfit to distinguish a single object.—It is only by a gradual transition from darkness to light that we are able to accustom our eyes to the brilliancy of noon.

§ 23. There is another peculiarity in the situation of our Earth with regard to the Sun, which you have not yet learned. The Earth's Axis is not even (parallel) with that of the Sun; but is somewhat inclined towards it, as represented in the last Diagram. To this is owing the _Change of the Seasons_. For on account of the inclination of the Earth's Axis, the Sun's rays fall, sometimes nearly perpendicular upon us, while at other times they are striking us more obliquely[5]. This is the principal cause of those

changes of temperature which we are in the habit of distinguishing by the names, _Spring_, _Summer_, _Autumn_ and _Winter_.

In winter the Sun’s rays strike us most obliquely; it is therefore the coldest season of the year. In summer they are most perpendicular;—Summer therefore is the hottest season. Spring and autumn are standing in the middle between these two. From spring till mid-summer the Sun’s rays are striking us more and more perpendicularly; from mid-summer till winter more and more obliquely.

A similar change of temperature is felt every day from Sun-rise (when the Sun-beams are most oblique) till noon, (when they are most perpendicular); and from noon again towards evening or Sun-set, when they are again oblique.

§ 24. It has been mentioned (Lesson IV, § 20), that each Planet in our Solar System is regularly turning on its Axis. But all of them do not perform this rotation with equal _velocity_. The Planets which are farther from the Sun are turning quicker than those which are near him. Jupiter, for instance, turns on its Axis twice as fast as our Earth. The nights in Jupiter, therefore, do not last half as long as ours.

This is, in some degree, necessary. For in proportion as a planet is further from the Sun, it receives less light and heat, which deficiency is, in part, made up by a more frequent exposition to his rays.

§ 25. The _Sun_ himself is also known to turn regularly on his Axis, and to complete one whole rotation in about Twentysix of our days. This we have been able to perceive from the spots which have been discovered on its surface, and which gradually move toward and disappear on one side, when in a short time after they appear again on the other.

§ 26. The Moon, and the Satellites of the other Planets have no rotary motion; but have always the same side turned towards their Planets. Thus the moon keeps constantly the same side turned towards the Earth; but her monthly motion round the Earth (Lesson III, § 17) is equal to a rotation on her axis; because by this means every part of her is, at least, once every Twentyseven days turned toward the Sun; as you may see from the Moon’s phases, represented on plate VII. A day in the Moon, therefore, is equal to Twentyseven of our days; because the Moon moves in Twentyseven of our days round the Earth, which is equal to turning once on her axis.

RECAPITULATION OF LESSON IV.

QUESTIONS.

[§ 20.] Is the progressive motion of the Planets round the Sun the only one which they are performing? What other motion have they besides this? What is this motion called? Why is it called so? What do you call a straight line imagined to be drawn through the centre of a Planet?

To what may the double motion of the Planets’ progression and rotary be compared?

[§ 21.] What time does the Earth need to turn once on its axis? How many times does the Earth turn on its axis during its whole journey round the Sun?[6] What is

this rotary motion of our Earth the cause of? Which portion of the Earth has day? Which night?

What would be the case, if the Earth was always to keep the same relative position with regard to the Sun? What is the advantage derived from the rotation of the Earth on its Axis. Would the Earth be habitable without it or not?

[The pupil ought now to explain plate IX.]

[§ 22.] What is the rotation of the Earth on its axis further the cause of? What does the gradual motion of each portion of our Earth towards the Sun produce?

What would be the case, if from the darkness of night we were at once exposed to the luminous rays of the Sun? Could our eyes endure the brightness of noon without a _gradual_ transition from darkness to light?

[§ 23.] What other peculiarity is there in the situation of the Earth with regard to the Sun? What is this the cause of? Do the rays of the Sun strike us at all times equally perpendicular or obliquely? What changes of temperature are thereby created?

How do the rays of the Sun strike us in winter? What season, therefore, is it? How do the rays of the Sun strike us in summer? What season, therefore, is summer with regard to temperature? How are the rays of the Sun striking us from spring till mid-summer? How, from mid-summer till winter? What similar change of temperature do we experience every day from Sun-rise till noon, and from noon till evening or Sun-set?

[§ 24.] Do all Planets turn on their axes with equal velocities? Which Planets turn quicker, those which are nearer or those which are further from the Sun? Give an instance. How long are Jupiter’s nights in comparison to our’s?

Why is this, in some degree, necessary?

[§ 25.] Is the Sun himself also known to turn on its axis like the Planets? How many days does he need for one complete rotation? By what means have we been able to observe this motion?

[§ 26.] Have the Moon and the Satellites of the other Planets also a _rotary_ motion? In what position does the Moon remain with regard to the Earth? But what is her monthly revolution round the Earth equal to? Why? What is a day in the Moon equal to? Why?

FOOTNOTES:

[4] The propriety of this comparison becomes still more evident, when we reflect that the Axis of the top is generally somewhat inclined to the plane.

[5] If the terms _perpendicular_ and _oblique_, should not be perfectly understood by the pupils, it will be easy for the teacher to explain their meaning.

[6] If the Earth requires 365 days to travel round the Sun; and each day has twentyfour hours; then the Earth will, during the whole of this time, turn Three Hundred and Sixtyfive times on its axis.

LESSON V.

OF THE APPEARANCE AND PECULIARITIES OF THE MOON AND SOME OF THE PLANETS WHEN VIEWED THROUGH A TELESCOPE.

§ 27. Next to the Sun there is no heavenly body so interesting to us as the _Moon_. When viewed through a good Telescope[7] she has nearly the same appearance as in Figure I, Plate X. The bright parts are supposed to be lofty mountains and tracts of land; (which is evident also from the shadow which they cast) and the dark spots are supposed to be valleys and caverns. Many of the mountains of the Moon are higher than the largest mountains on the Earth. Some of them are volcanos, and their eruptions have been distinctly observed by many distinguished Philosophers. Some of the caverns are ascertained to have a depth of many miles and a width of almost Three miles.—No water has as yet been discovered in the Moon. Hence if she is inhabited, as we have reason to believe, her inhabitants must be very differently constructed from ourselves.

§ 28. Among the Planets _Venus_ is by far the most beautiful in appearance. She is known also by the name of the _Morning_ and _Evening Star_. Her light is so bright that she is often seen at _Noon_. When viewed through a good telescope she exhibits phases similar to those of the Moon (Lesson III, § 17), which proves her spherical form (Lesson I, § 3). The mountains in Venus have been calculated to be at least Six times as high as those on our Earth. Her Atmosphere is only half as dense as ours.

§ 29. Mars appears in many respects similar to our Earth. His light is red and changeable; his surface exhibits black changeable spots (see Figure II, Plate X). Some philosophers pretend to have noticed a region of ice on his poles. His atmosphere is twice as dense as ours.

[Illustration:

_N^o. X._]

§ 30. Jupiter, viewed through a telescope, exhibits a surface covered with stripes. These are supposed to be clouds. A representation of them is given in Fig. III, Plate X. His light is very white and subject to but little variation. His atmosphere is nearly Twentyseven times denser (thicker) than ours.

§ 31. Very remarkable, as we have already observed, is the Planet Saturn, on account of its luminous ring. Viewed through a telescope it has the appearance, represented in Figure IV. It is highly probable that to the inhabitants of that Planet, this ring has an entirely different appearance from what it has to us. It appears to be a solid opaque mass, and is probably inhabited like the Planet, which it constantly accompanies on its journey round the Sun. Saturn’s atmosphere is nearly Ninety times denser than that of our Earth. No mountains have as yet been discovered on its surface.

* * * * *

Herschel is too remote, for us to know much about its surface. Its atmosphere is supposed to be Three Hundred and Sixty-one times thicker than ours.

Mercury, being the nearest Planet to the Sun, has a very bright light; but is only seen early before Sun-rise, and immediately after Sun-set. It exhibits Phases like the Moon.

RECAPITULATION OF LESSON V.

QUESTIONS.

[§ 27.] What do you know about the Moon's surface? What, about its mountains, volcanos and caverns? Have any great waters been discovered in it?

[§ 28.] What do you know about the Planet Venus? By what other name is she known? What does she exhibit, when viewed through a telescope? What do you know about her mountains? What, about her atmosphere?

[§ 29.] What do you know about the Planet Mars? What, about his light, surface and atmosphere?

[§ 30.] What surface does Jupiter exhibit when viewed through a telescope? Of what color is the light of Jupiter? What do you know about his atmosphere?

[§ 31.] What do you know about the Planet Saturn? What does the ring of Saturn appear to be? What do you know about the atmosphere of this Planet?

FOOTNOTES:

[7] This is an instrument through which we can see things much clearer and larger than we could with the naked eye.

LESSON VI.

DESCRIPTION OF THE EARTH'S SURFACE.—DIVISION OF IT INTO ZONES.—TORRID, TEMPERATE AND ARCTIC ZONES.—THE FIVE GREAT CONTINENTS.—RELATIVE DIMENSIONS OF AMERICA, EUROPE, ASIA, AFRICA AND AUSTRALIA.

§ 32. Having learned the relation of our Earth to the Sun and the Planets, it will be well to acquaint ourselves with the principal objects on its surfaces.—What the _Interior_ of our Earth consists of, we have as yet no idea, because we have not yet been able to penetrate deeper than a few hundred feet; and this is, in proportion to the Earth's Diameter, little more than nothing. But as regards her surface, we know that it consists partly of land and partly of water. Little more than one fourth of the Earth's surface is covered with land; all the rest is water.—The great lands are called _Continents_; the great waters are called _Oceans_. Smaller portions of land surrounded on all sides by water, are called _Islands_. Smaller bodies of water surrounded by land are called _Lakes_.

§ 33. The land on our Earth is divided into Five Continents: America, Europe, Asia, Africa and Australia.—These, however, are not all of the same extent. Europe is the smallest of them. America and Asia are the greatest.

The adjoining Plate No. XI, will give you an idea of the proportion of land and water on our Earth, and of the relative extent of the Five Continents.

Fig. I represents the surface of the Earth divided into land and water. Were all land on our Earth put together in a circle, and the water placed round it, then the land would only fill the inner circle, the water occupying the surrounding ring, a space nearly four times as large as the circle. Fig. II, III, IV, V and VI represent the comparative surfaces of America, Europe, Asia, Africa and Australia. Fig. VII represents the comparative surface of the Moon.

Upon close inspection of these figures you will perceive;

1. That the extent of America and Asia are nearly equal; but that each of these Continents is several times larger than either Europe or Australia.

2. That the next greatest portion of our Globe is _Africa_, which is more than three times larger than Europe.

3. That Europe is the smallest Continent of our globe.

4. That the whole surface of the Moon would not be more than enough to cover either America or Asia.

[Illustration:

_N^o. XI._

Fig. 1. THE OCEAN THE LAND

Fig. 2. EUROPE

Fig. 3. ASIA

Fig. 4. AFRICA

Fig. 5. AMERICA

Fig. 6. AUSTRALIA

Fig 7. THE MOON’S SURFACE

Geographical Miles.]

[Illustration:

N°. XII.

Fig. 1. THE WHOLE TORRID ZONE

Comparative Extent of the TWO TEMPERATE _ZONES_

Comparative EXTENT _of the_ TWO ARCTIC ZONES

Fig. 2.

Comparative Extent of one of the WHOLE TORRID _ZONE_

Fig. 3.

Comparative Extent _of one of the TWO TEMPERATE_ ZONES

Fig. 4.

ONE _of the ARCTIC_ ZONES

Fig. 5.

Comparative Extent of the WHOLE LAND on our GLOBE]

[Illustration:

N°. XIII.]

[Illustration:

N°. XIV.

Square contents of the Two TEMPERATE ZONES.

Square contents of the WHOLE TORRID ZONE.

Square contents of the Two ARCTIC ZONES.

The whole surface _of the five GREAT CONTINENTS on our_ GLOBE _Compared to one of the two_ TEMPERATE ZONES.]

§ 34. The Earth’s surface is not throughout equally illumined or heated by the Sun; because the Sun’s rays strike some portions of the Earth more perpendicular than others. Our Earth, therefore, is divided into _Climates_ or _Zones_, which you will understand better from Plate XII.

You will see from it that the Sun’s rays are perpendicular to the central part of the Earth’s surface; but that toward the extremities of the Earth’s Diameter these rays strike us more and more _obliquely_. The greatest heat, therefore, must be felt by the people living between the two circles EF and IK. The circle GH, which is exposed to the perpendicular rays, is termed the _Equator_; and the two circles EF and IK, _which are at equal distance from the Equator, are called Tropic Circles_. The whole surface included by these two circles is called the _torrid Zone_. The space between either of the circles CD and EF, or IK and LM, is called a _temperate Zone_; because the Sun’s rays striking these portions neither perpendicular nor very obliquely, their inhabitants suffer neither great heat nor cold. In one of these Zones are situated the United States of America and the greater portion of Europe. Beyond them, toward the extremities of the Diameter AB, are the two _icy_ or _arctic_ zones. The Sun’s rays strike them very obliquely; which is the cause of their being almost continually covered with ice or snow.

The two circles, CD and LM, are called _Polar_ circles; and the two extremities, A and B, of the Earth’s Diameter, situated in those regions, are called _the Poles_. A is called the _North-Pole_ and B the _South-Pole_ of the Earth.

§ 35. The different zones of which we have just spoken, are not equal to one another. Plate XIII, will show their relative extent.

Fig. I represents the surface of the Earth divided proportionally into its three zones: the _torrid_, the _temperate_ and the _arctic_. The inner circle represents both the arctic zones; the yellow ring b, which surrounds it, represents the united extent of the two temperate zones; and the outmost red ring, the whole of the two _torrid zones_.

Fig. II represents separately the whole torrid zone;—Fig. III one of the temperate zones;—Fig. IV one of the arctic zones; and Fig. V the whole extent of land on our globe.

The next Plate, No. XIV, represents the comparative surfaces of these zones, drawn separately in form of squares; and the last figure on that Plate, shows the extent of the five continents, compared to one of the temperate zones.

[Illustration:

_N^o. XV._

Fig. 1. _Fig. 2._ _Fig. 3._ _Fig. 4._ _Fig. 5._ _Fig. 6._]

You will observe from a close inspection of these figures, that the whole extent of land on our globe is nearly equal to that of a temperate zone; and that if it were possible to unite America, Europe, Asia, Africa and Australia into one, their united extent would not yet fill one of the temperate zones! You will also perceive that the two temperate zones occupy together the greatest portion of the Earth’s surface, and that the arctic zones occupy comparatively the smallest.

RECAPITULATION OF LESSON VI.

QUESTIONS.

[§ 32.] Do we know anything about the Interior of our Earth? Why not? What does the surface of our Earth consist of? What proportion does the land bear to the water? What are the great lands called? What, the great waters? What are smaller portions of land, surrounded by water, called? What, small portions of water, surrounded by land?

[§ 33.] Into how many continents is all the land of our Globe divided? What are they? Are all the continents of our globe of the same extent? Which is the smallest of them? Which are the largest? Explain Plate, No. XI.

What proportion does the extent of America bear to that of Asia? What relation do these continents bear to Europe or Australia? What is the next greatest portion on our globe? Which continent is the smallest? What relation does the surface of the Moon bear to America or Asia?

[§ 34.] Is the whole Earth equally illumined or heated by the Sun? Why not? What is, therefore, the surface of our Earth divided into? Upon what portion of our Earth do the Sun’s rays fall perpendicular? What portion do they strike more obliquely? What people, therefore, will experience the greatest heat?

[The pupil ought now to explain Plate XII. The elder pupils ought to draw a sphere with the Equator, the tropic and arctic circles. They ought also to draw the Diameter of the Earth, and indicate the North and South Pole.]

[§ 35.] If you compare the whole extent of land on our globe to the Contents of one of the temperate zones, what proportion do you find them to bear to each other? If it were possible to unite the five great continents, America, Europe, Asia, Africa and Australia, what zone would they nearly fill? What two zones occupy the largest portion of the Earth's surface? What two, the smallest?

APPENDIX

CONTAINING THE COMPARATIVE POPULATION OF THE DIFFERENT QUARTERS OF OUR GLOBE.

[The numbers are given in Table III.]

§ 36. The Five principal parts of our globe, America, Europe, Asia, Africa and Australia are not equally thickly settled. Europe and Asia have, in proportion to their extent, the greatest population; America and Australia the least.—The following Plate, No. XV, shows the comparative population of these continents.

Fig. I represents the whole surface of land on our globe, inhabited by nearly One Thousand Millions (One Billion) human beings. If these were to live throughout as close together as in _Europe_, then they would only occupy a surface of land as large in proportion, as the inner circle marked _a_. But the two rings, _b_ and _c_, occupy each as much surface as the circle _a_; hence there is yet room for twice as many human beings; before each quarter of the world is as thickly settled as Europe.

Fig. II represents _Asia_ and its population. If this quarter were settled as thickly as Europe is, then its inhabitants would only fill the inner circle marked _b_; the ring _a_, therefore, is still left for settlement.

Fig. III exhibits the population of Africa. If the inhabitants of this continent lived as close together as those of Europe, they would only fill the inner circle, marked _c_, and the surrounding ring might yet be inhabited.

Fig. IV shows the comparative population of America. Its inhabitants, crowded together as the inhabitants of Europe, would only occupy the small circle _e_; the whole broad ring _f_, therefore, is still left for settlement!!

Fig. V represents Australia. Its inhabitants, settled as in Europe, would only fill the circle _a_.

Fig. VI represents the population of Europe filling the whole of that Quarter.

The whole of these Six figures may represent to the pupil the comparative extents of the five great continents of our globe; but the _inner circles_ of these figures, and the whole of the sixth figure, show their comparative populations. From a close inspection of this plate the pupil may learn:

1. That the population of Asia is yet greater than that of all the rest of the world. (The circle _b_ in figure II being yet larger than the inner circles of all the other figures, and figure VI taken together.)

2. That the population of Europe is as yet larger than that of America, Africa and Australia, taken together.

3. That the population of Africa is larger than the joint populations of America and Australia.

4. That America if once settled as Europe is, will have more than Six times her population.

[The teacher, if he think proper to ask the pupils some questions in reference to the Appendix, will find no difficulty in adapting them to the capacity of his pupils.]

TABLE I.

Showing the Diameter, Surface, and Cubic Contents of the Sun and the Planets.

Diameter in Surface in Cubic Contents Geographical Geographical in Geographical Names. Miles. Square Miles. Cubic Miles. SUN, 194,000 118,093,000,000 3,825,903,253,970,000 Mercury, 608 1,161,314 117,659,099 Venus, 1678 8,844,063 2,473,469,743 Earth, 1719 9,282,066 2,659,159,061 Mars, 1006 3,178,805 532,996,317 Vesta, 74 15,000 2,121,347 Juno, 309 282,690 2,355,750 Ceres, 352 389,182 22,832,034 Pallas, 465 650,266 52,886,472 Jupiter, 19566 1,202,280,406 23,533,143,597,631 Saturn, 17263 936,530,620 2,757,547,946,775 Herschel, 7564 173,696,911 1,359,227,438,858 Moon, 480 723,686 51,561,578

TABLE II.

Showing the exact Duration of the Revolutions of the different Planets round the Sun.

Duration of the Planets. Years. Days. Hours. Min. Sec. Moon’s revolution round the Earth. Mercury, — 87 23 15 44 27 days, 7 hours, 43 Venus, — 224 16 49 10 minutes, and 12 Earth, 1 — 6 9 8 seconds. Mars, 1 321 22 18 31 Vesta, 3 225 — — — Juno, 4 131 10 30 — Ceres, 4 220 13 4 — Pallas, 4 221 15 35 — Jupiter, 11 314 20 39 — Saturn, 29 166 2 — — Herschel, 83 266 9 — —

TABLE III.

Showing the Extent and Population of the five great Continents.

Names of the Continents. Extent in Sq. Miles. Population. America, 14,868,000 40,000,000 Europe, 3,292,000 198,000,000 Asia, 15,000,000 500,000,000 Africa, 11,267,900 150,000,000 Australia, 3,823,200 1,500,000

The United States, 1,781,926 13,000,000

POPULAR LESSONS IN ASTRONOMY, ON A NEW PLAN;

IN WHICH

SOME OF THE LEADING PRINCIPLES OF THE SCIENCE ARE ILLUSTRATED BY ACTUAL COMPARISONS, INDEPENDENT OF THE USE OF NUMBERS.

BY FRANCIS J. GRUND,

AUTHOR OF “AN ELEMENTARY TREATISE ON PLANE AND SOLID GEOMETRY,” “ELEMENTS OF NATURAL PHILOSOPHY AND CHEMISTRY,” &c. &c.

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